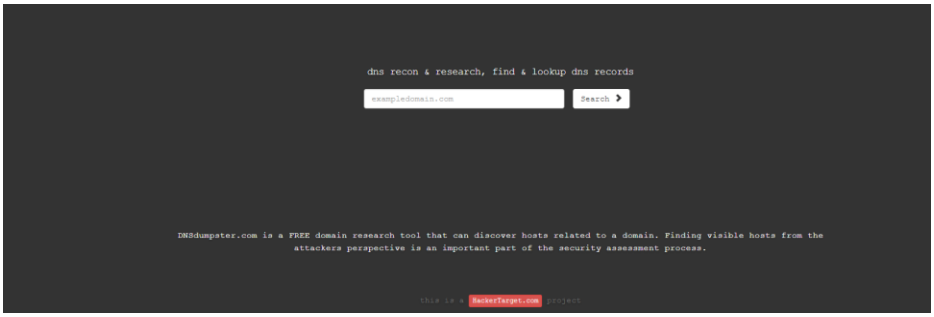
What are MX records?

Mail Exchange records are an integral part of the Domain Name System (DNS) that plays a crucial role in routing email messages. Located in DNS zone files, which are simple text files bundling all records for a particular domain, MX records provide email clients with information about the domain under which a mail server can be accessed. Using the Simple Mail Transfer Protocol (SMTP), they are responsible for directing emails to the correct mail server.

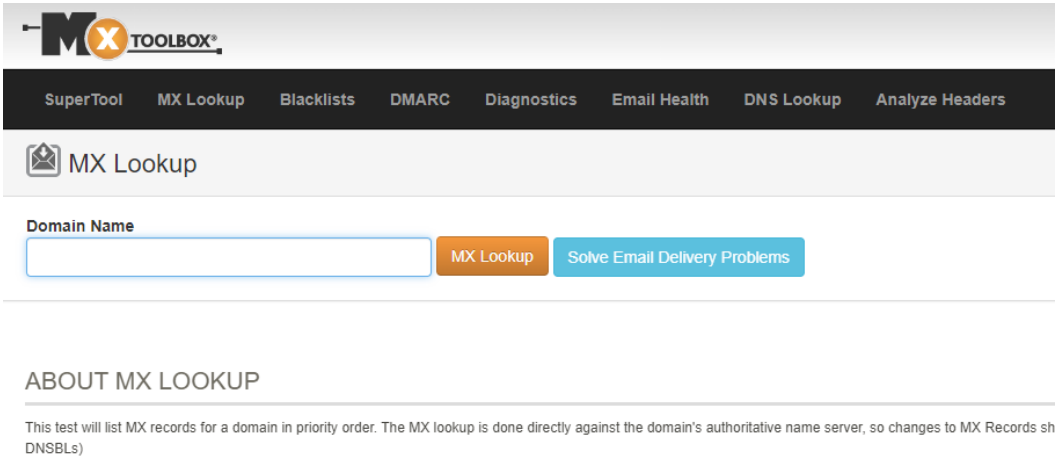
Within a domain, there are usually multiple servers, including web, File Transfer Protocol (FTP), and one or more mail servers, each accessible via different subdomains managed by DNS records. Being a type of DNS record, an MX record enables clients to request the name of the subdomain associated with the mail server, which manages the sending and receiving of emails.

Websites you can use:

https://dnsdumpster.com/



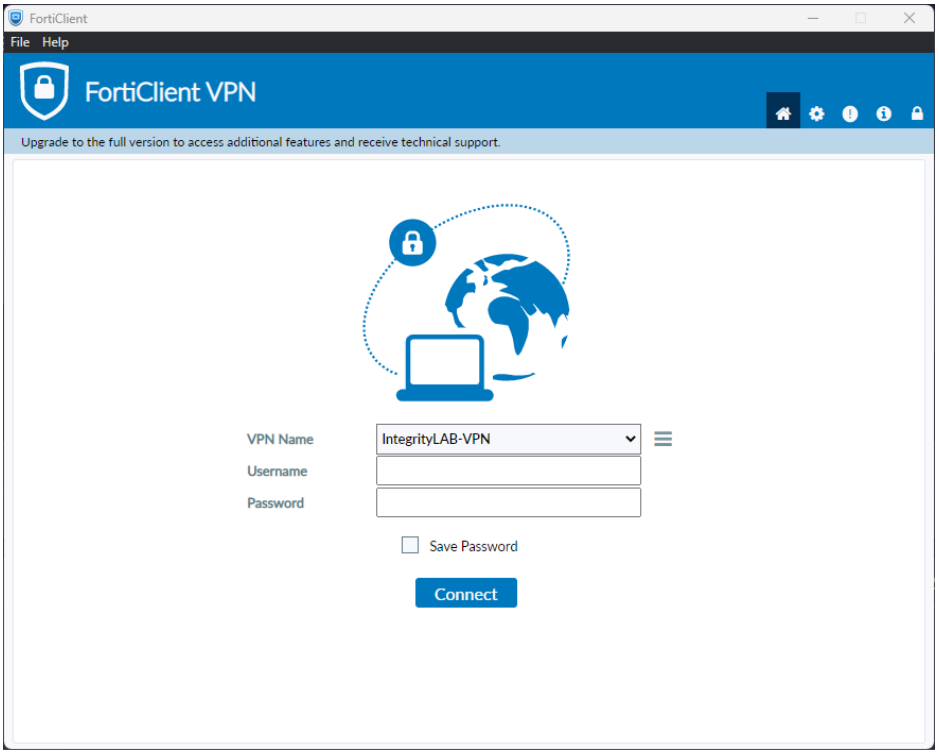
https://mxtoolbox.com/



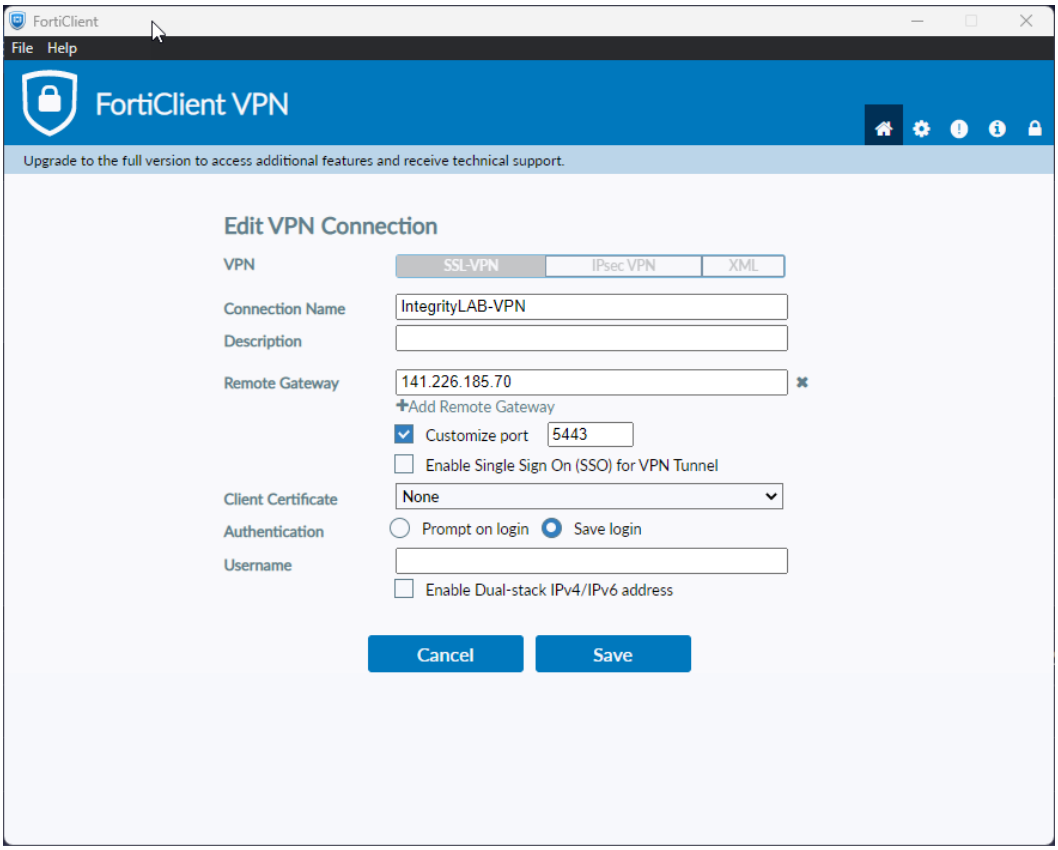
How to start MX Record Check:

First you need to have an account in the Forti client VPN for the Cyberintegrity Company. For that you need to contact the relevant people in the company to open an Account for you.

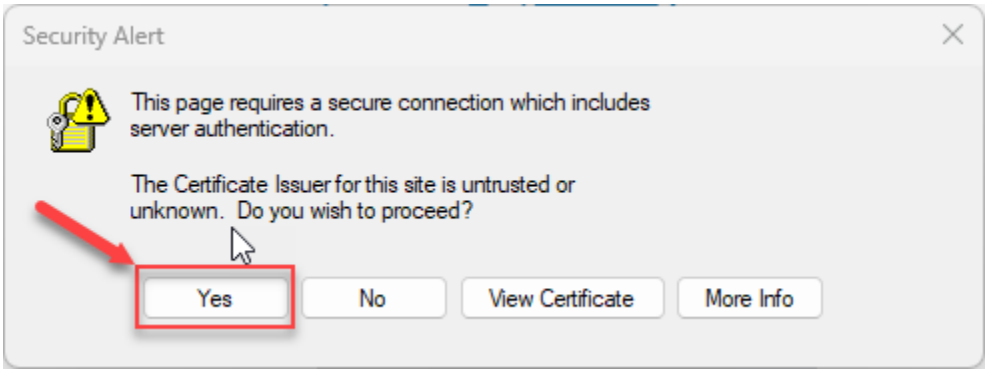
The FortiClient VPN looks like.



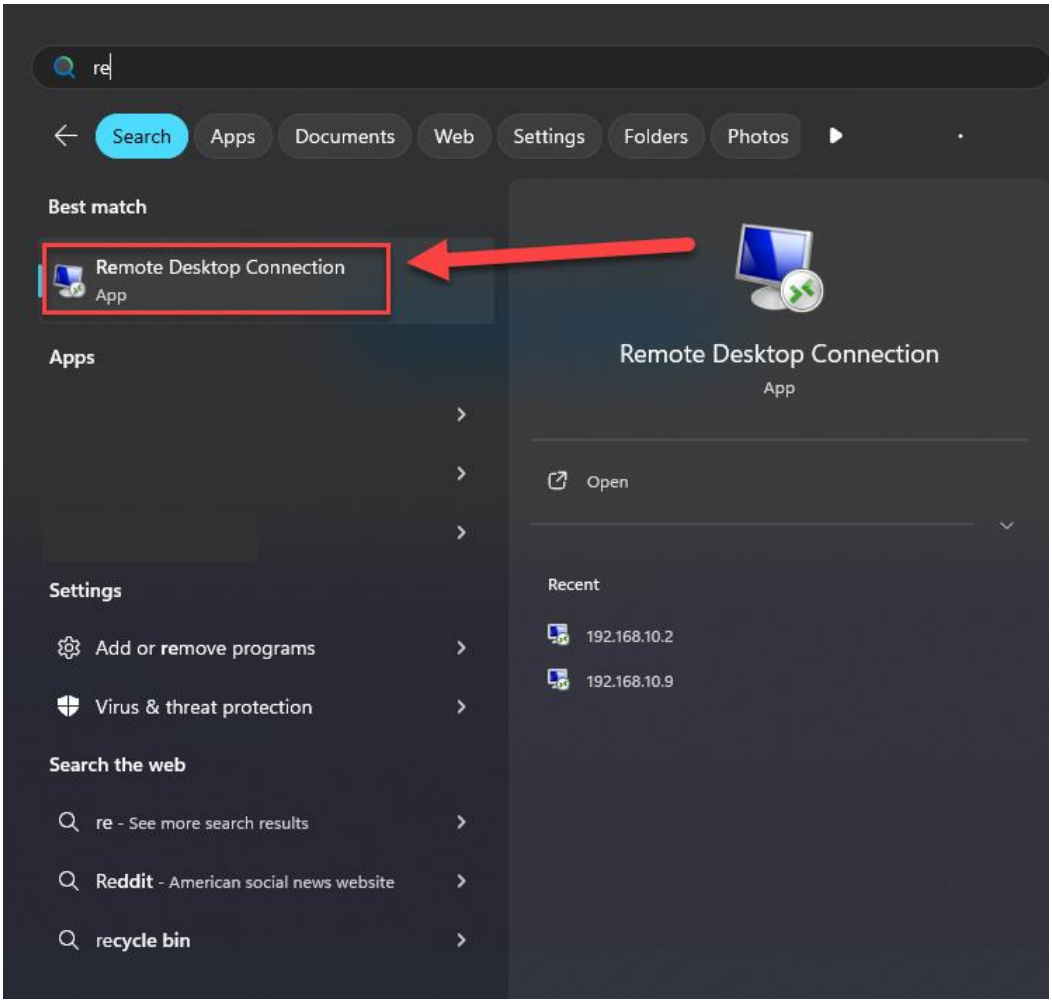
You need to configure the setting as the following:



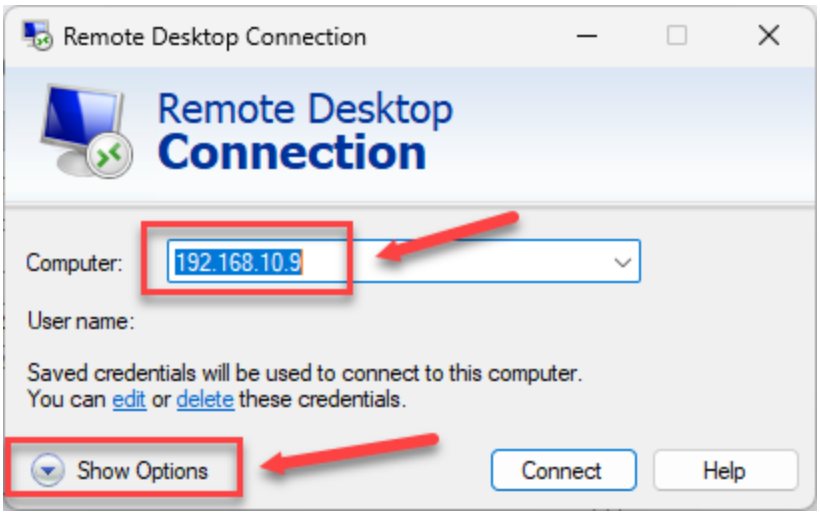
After you connect you should get an alert like shown below: you press yes.



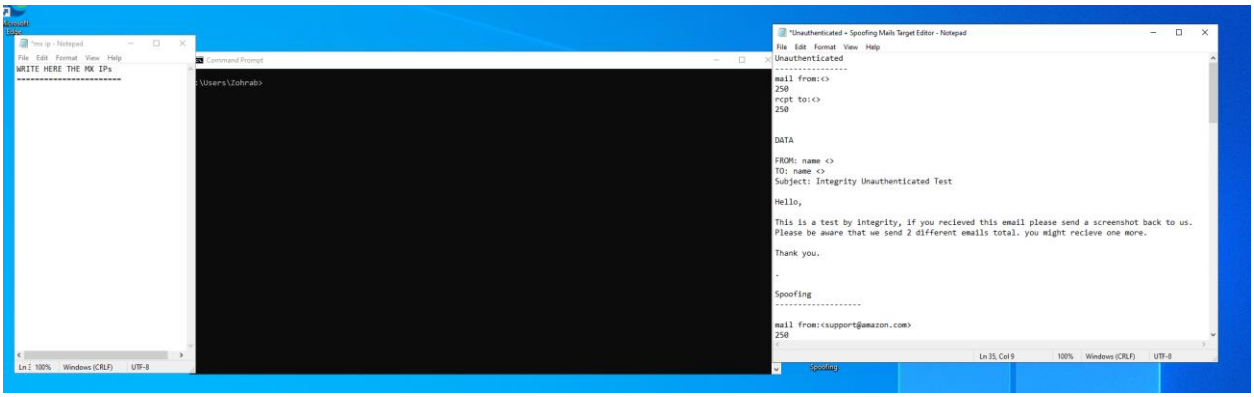
After the connection you will go to Remote Desktop Connection by searching in the search bar:



By clicking on Remote Desktop Connection a windows will be opened and you need to press on show options and then enter this IP address 192.168.10.9 and hit connect.

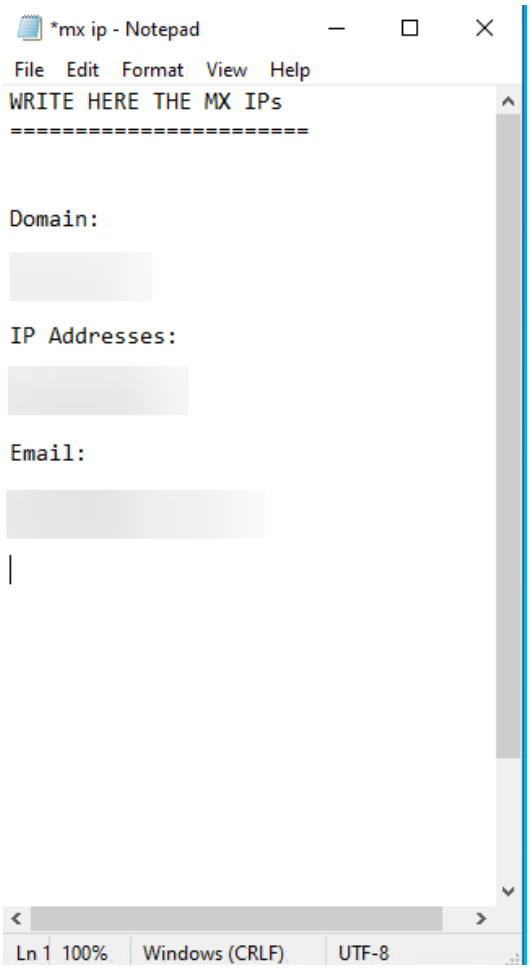


Once you get connected you will see this windows:



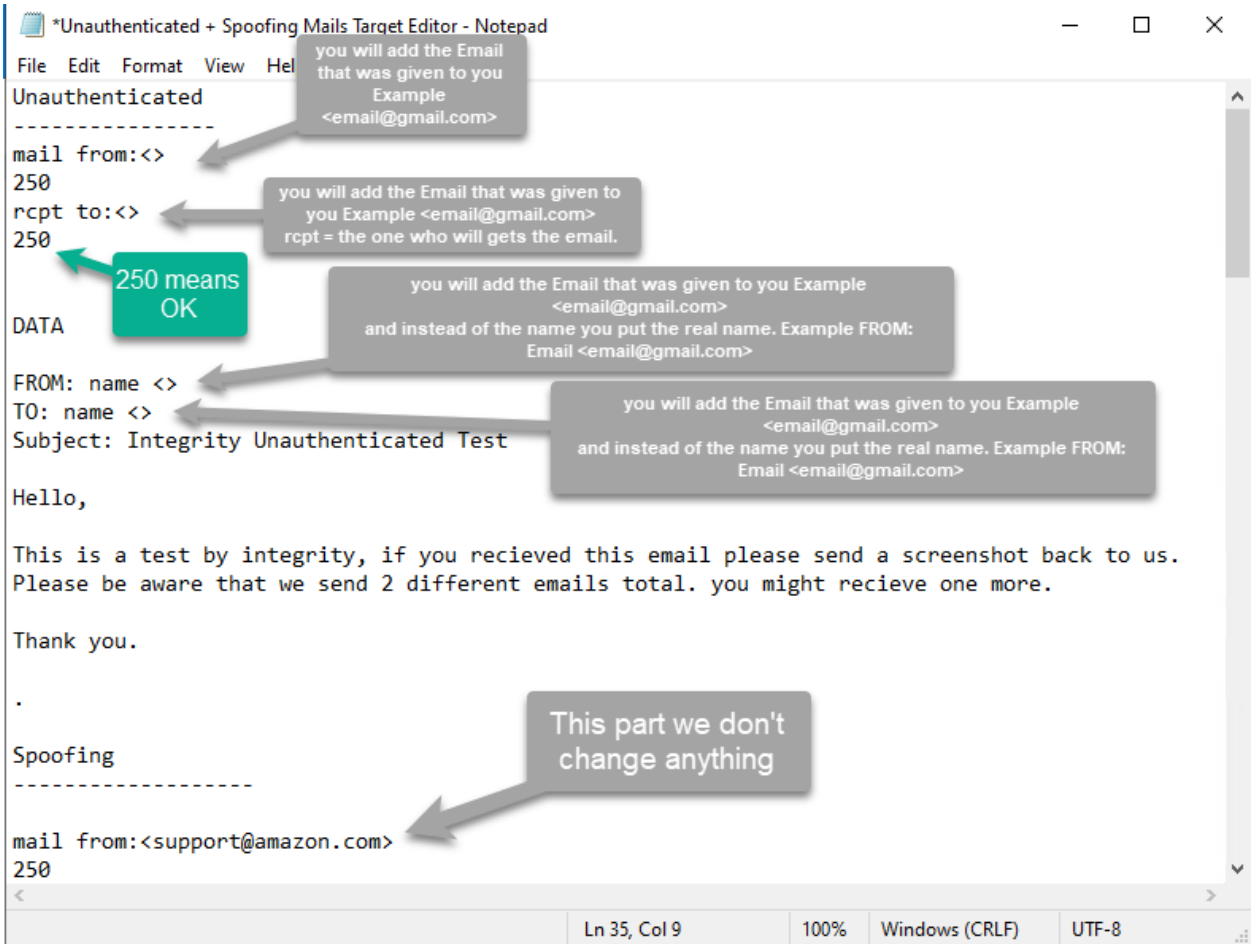
In the first section you will write:

- **Domain**
- **IP Address for the mail relay**
- **The Email**



The Second step will be filling the following:

Replace the "email@gmail.com" with the real email address and the Name with the real Name.



Unauthenticated
----
mail from:<Email@gmail.com>
250
rcpt to:<Email@gmail.com>
250

DATA

FROM: Email <Email@gmail.com>
TO: Email <Email@gmail.com>
Subject: Integrity Unauthenticated Test

Hello,

This is a test by integrity, if you recieved this email please send a screenshot back to us.
Please be aware that we send 2 different emails total. you might recieve one more.

Thank you.

.

For the Spoofing section you do the same as the first one:



Spoofing
----

mail from:<support@amazon.com>
250
rcpt to:<email@gmail.com>
250

DATA

FROM: Amazon Support <support@amazon.com>
TO: Email < email@gmail.com >
Subject: Integrity Spoofing Test

Hello,

This is a test by integrity, if you recieved this email please send a screenshot back to us.
Please be aware that we send 2 different emails total. you might recieve one more.

Thank you.

.

For the Mail-Relay Check section you do the same as the first and the second sections:



mail from:<Zohrab@cyberintegrity.co.il>
250
rcpt to:<Zohrab@cyberintegrity.co.il>
250

DATA

FROM: Zohrab <Zohrab@cyberintegrity.co.il>
TO: Zohrab <zohrab@cyberintegrity.co.il>
Subject: Integrity Mail Relay Test

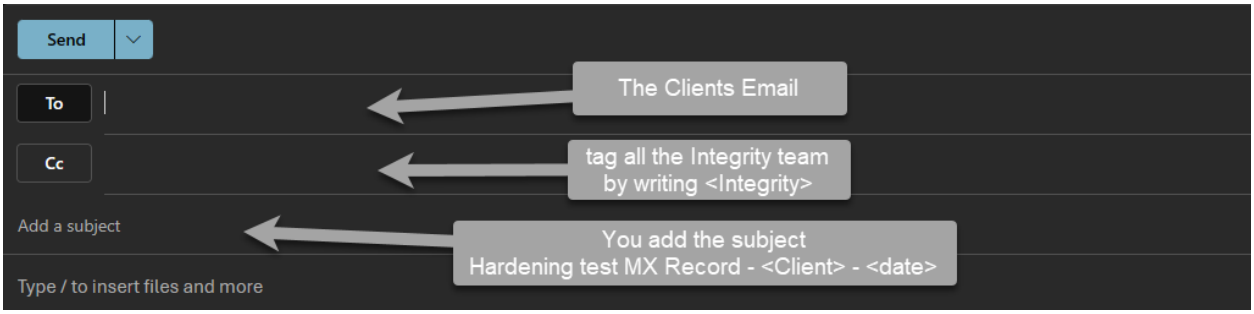
Hello,

This is a test by integrity.
If you recieved this Email then the MX that has been checked can be used as a mail relay.

Thank you.

.

Before Starting the MX Record Examination, you need to send an email to the client:



You'll need to write the following:

Greetings,
Soon I will test the MX servers by sending a spoofing email, pretending to be someone from the organization and checking the Mail relay, it is possible that you will receive an email from someone in the organization with a message from Integrity.
I'll send screenshots once I'm done,
Thank you.

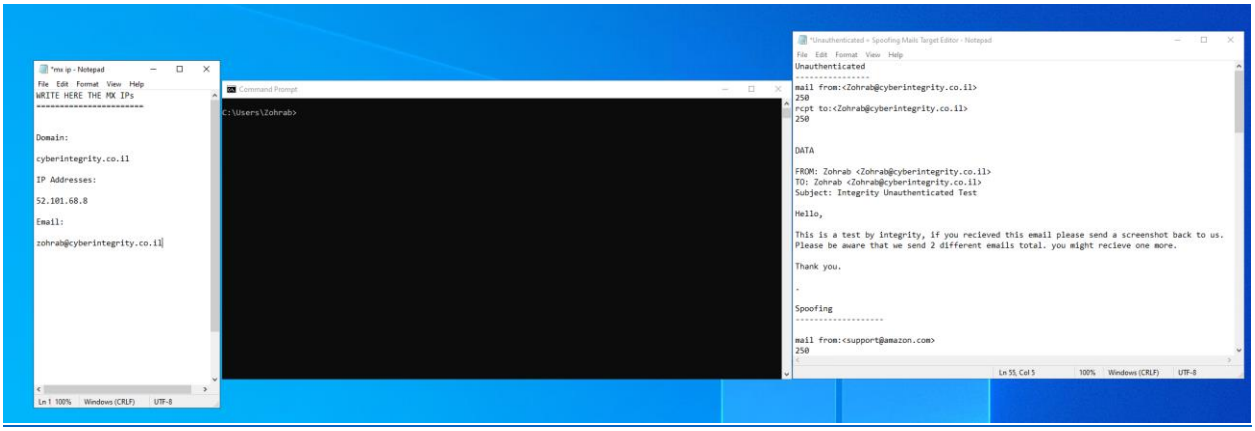
After sending this email you need to check the mail relay:
Need to go to https://dnsdumpster.com/ and search in the search bar the client's domain, for this example I'll be using cyberintegrity.co.il.



You need to go the section below and copy the IP addresses (if there are more than one) and paste it in the mx server in the notepad.

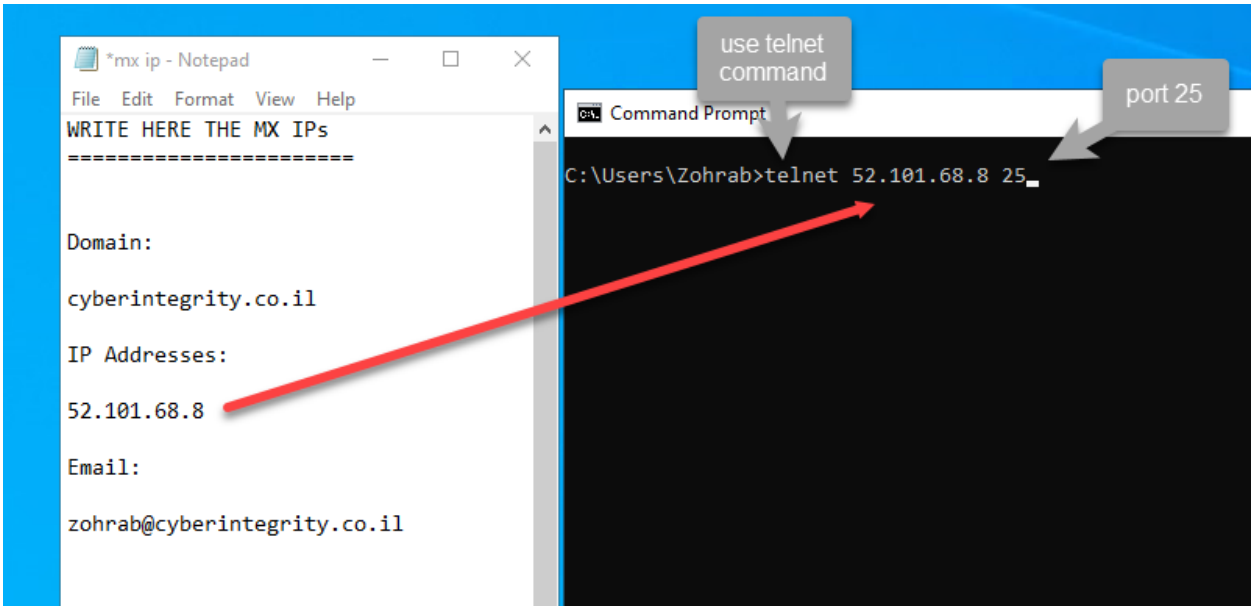


This is how the mx server should look like when you configure the credentials (for this report and example I'm using the domain: Cyberintegrity.co.il and the email: [zohrab@cybetintegrity.co.il)](mailto:zohrab@cybetintegrity.co.il)



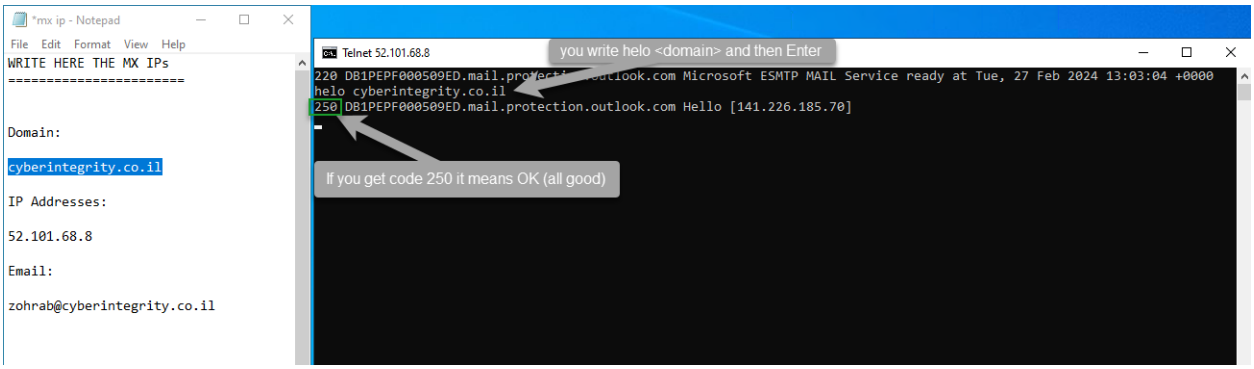
The second step is to use this command:

- **telnet <IP address> 25 and press enter.**

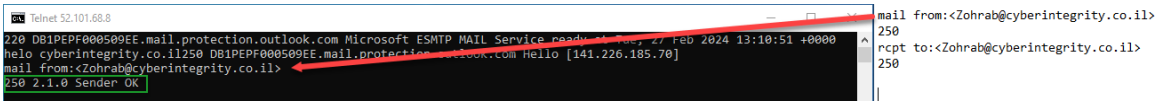


The third step is to write as the following:

Helo <domain> and then Enter.



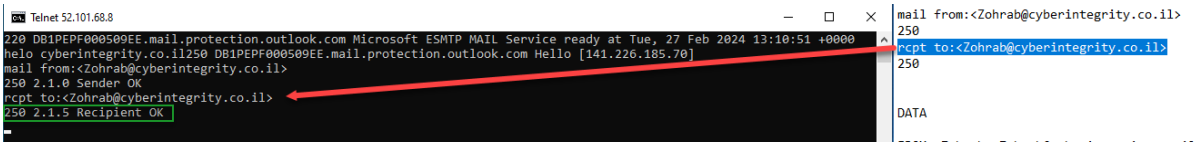
You enter first the mail from: <the real email in this example its zohrab@cyberintegrity.co.il>



If you get the code 250 then you can continue

Then

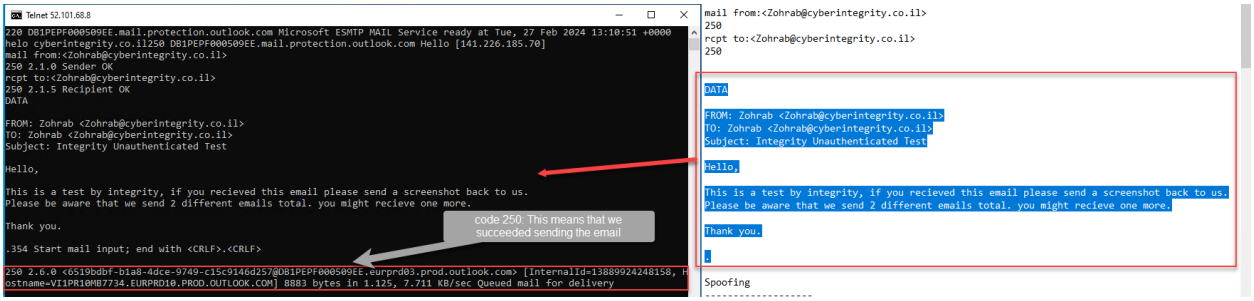
You enter the rcpt to: : <the real email in this example its zohrab@cyberintegrity.co.il>



If you get the code 250 then you can continue

Then

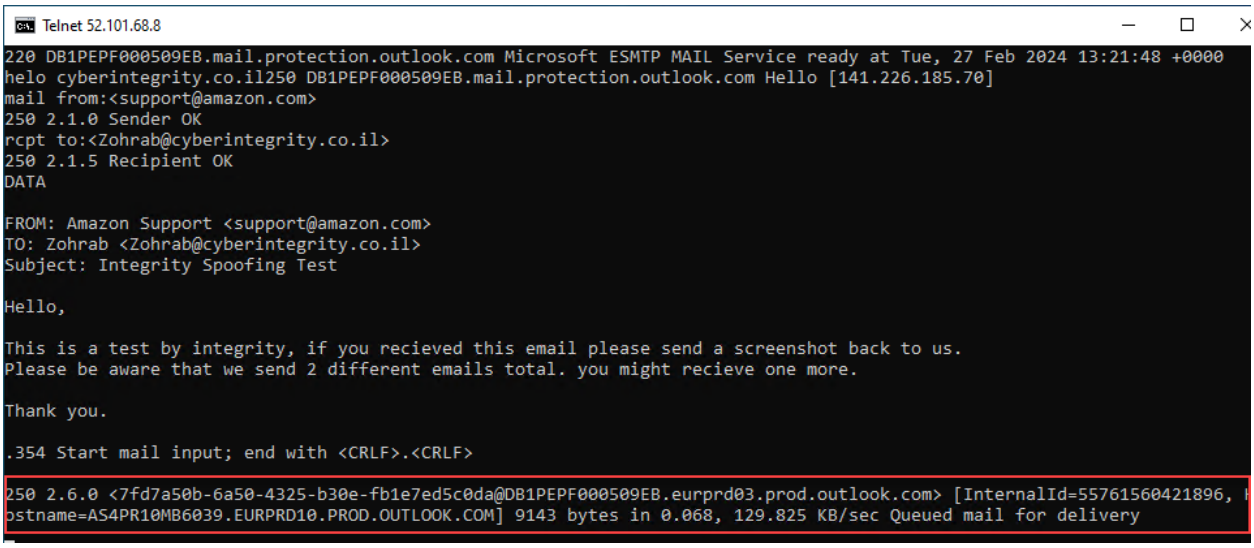
You copy the from the DATA to the dot <.> and press Enter:



Code 250: This means that we succeeded in sending the email, but it doesn't mean that the client got it, so the client needs to check if he got the email or not.

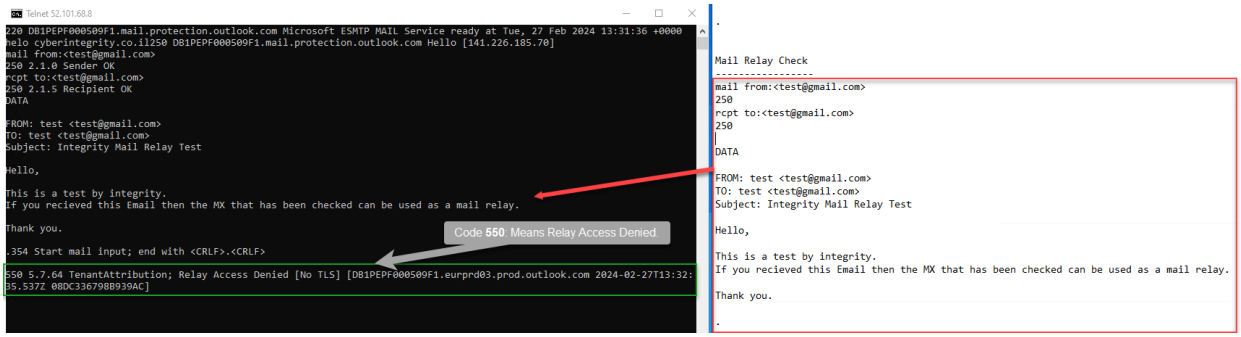
The same steps you do for the Spoofing Section:

This is the result:



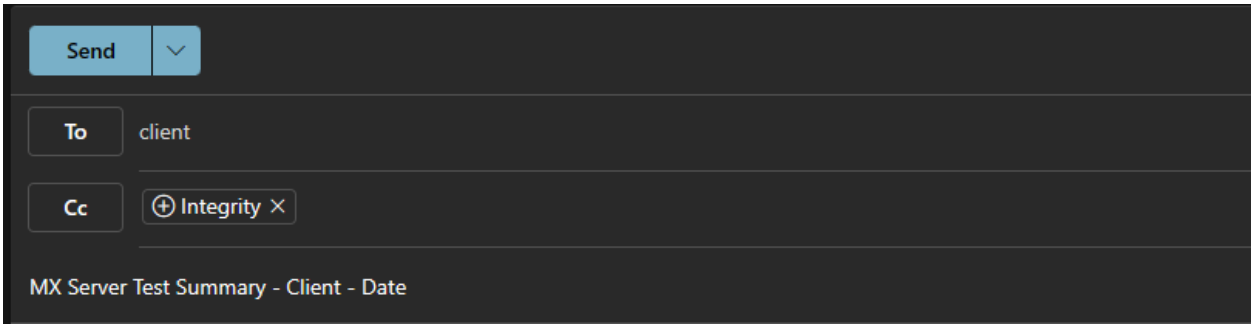
We also got code 250. This means that we succeeded in sending the email, but it doesn't mean that the client got it, so the client needs to check if he got the email or not.

The same Steps you do for the Mail-Relay section:
But in this section the mail from and the rcpt need to be the cyberintegrity
In this example I will be using the Gmail account "test@gmail.com":



In this example we got the Code 550 which means Relay Access Denied, which means you couldn't send the message, and its good.

Sending the final report to the client:



Then

Greetings,

Further to your approval to perform impersonation tests on the email servers, below is the summary of the test:

- The checked email domain: cyberintegrity.co.il
- MX Records addresses checked:

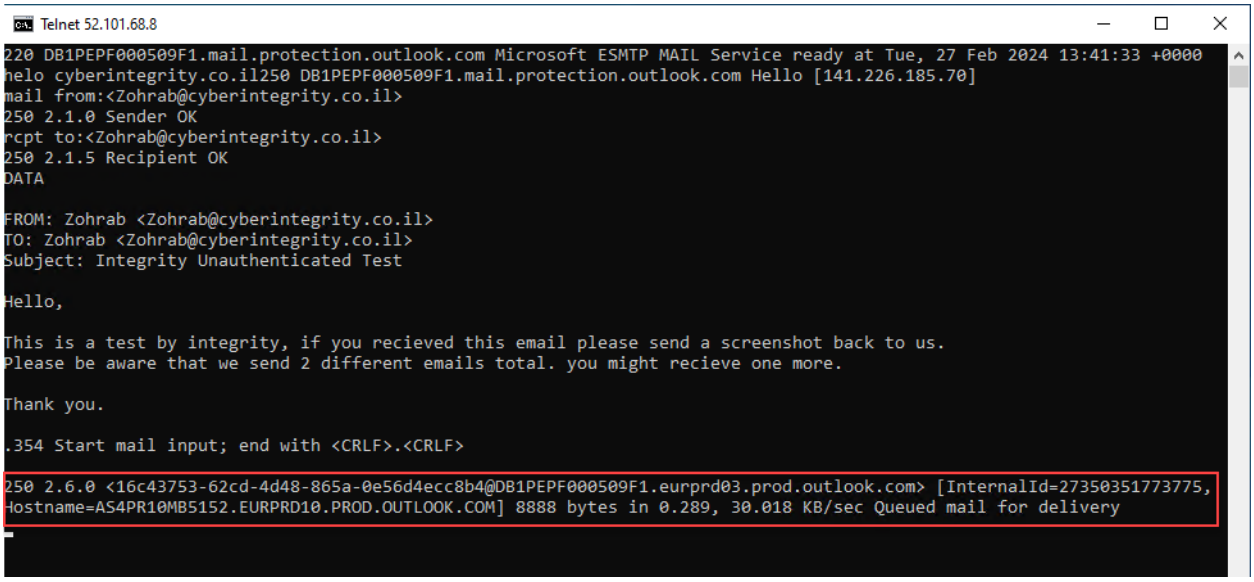


Here are the tests done on the MX address:

1. Impersonating an email from within the organization.
2. Email spoofing.
3. Mail Relay attempt on the MX server.

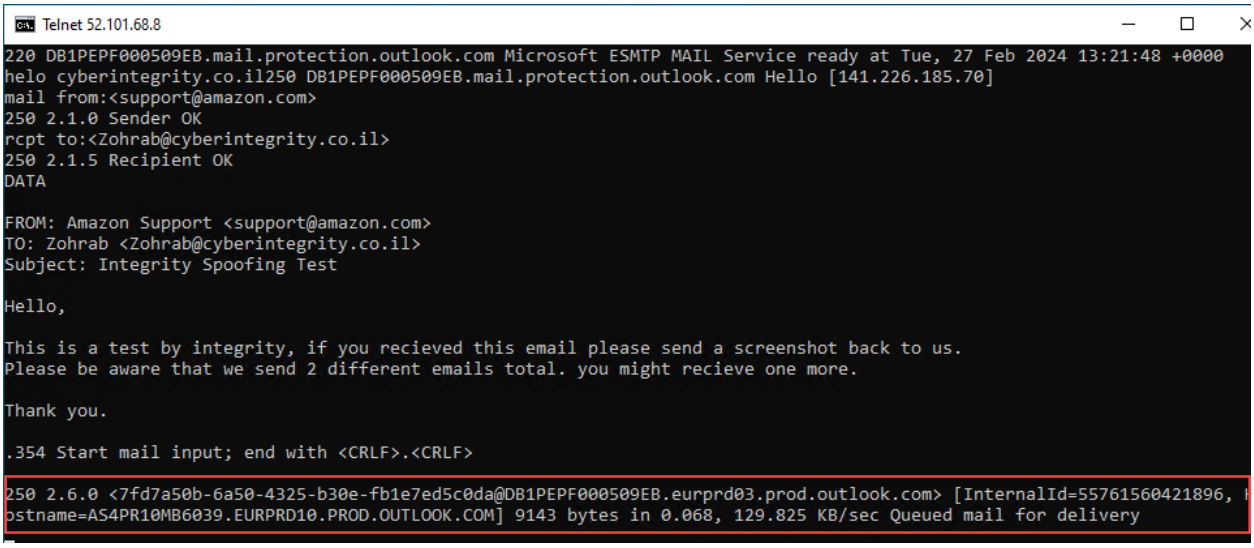
Impersonating an email from within the organization

As you can see in the picture, we **managed** to send an impersonation email to an email from within the organization:



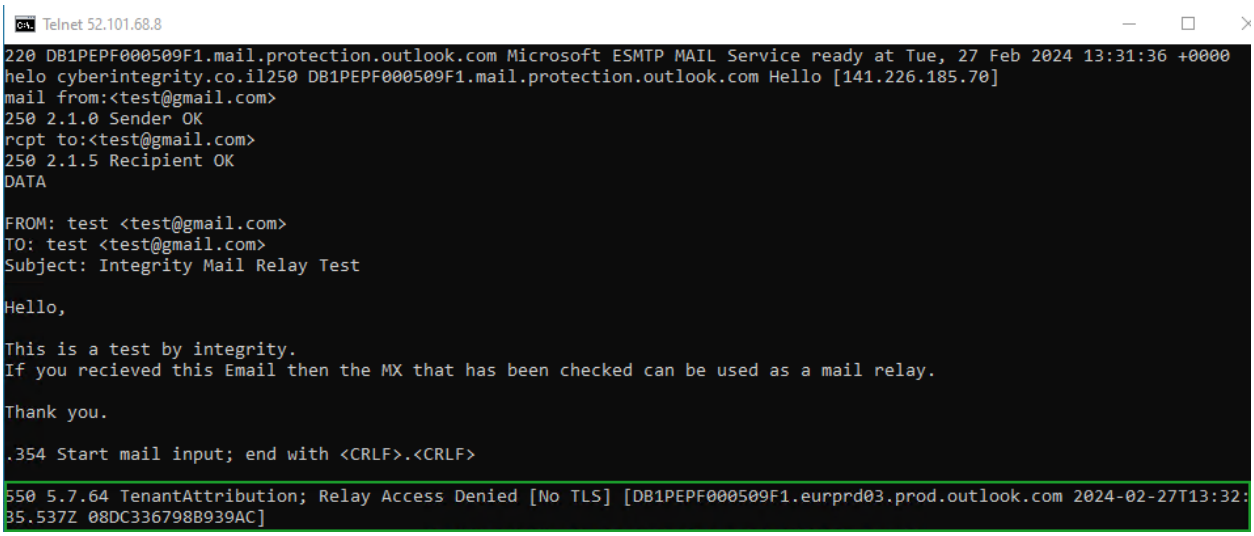
Email spoofing.

As you can see in the picture, we **managed** to send a spoofing email to the organization:



Mail-Relay attempt on the MX server

As you can see in the image, we **failed** to perform a Mail Relay attack



If you receive an email from the test we performed, I would appreciate it if you attached to this email a screenshot detailing which box it entered (Junk, Spam, MailBox).